

EMAIL SECURITY ANALYSIS

Phishing / Malicious Email Assessment

EPVME sample: 000b3003b95fb4a634cadb2fde61cc74.eml

Assessment	MALICIOUS — high confidence
Primary finding	Multiple From header / sender identity confusion
Dataset	EPVME-Dataset, EPVME_1.zip
Analyst scope	Static email/header, MIME, embedded object and domain/DNS review

1. Executive Summary

The email was analyzed based on its header structure, authentication information, MIME structure, message content, and embedded file. The main suspicious finding is the presence of two different email addresses in the **From** header, belonging to different domains. This structure is consistent with the “Multiple From Header” technique documented in the EPVME dataset.

The embedded **newspaper.gif** file was also analyzed separately. It was confirmed as a valid GIF89a image, and the static analysis performed did not identify an obvious malicious payload.

Based on the combined findings, the email is assessed as malicious/phishing-related. However, the available evidence does not prove a specific objective such as credential theft, malware delivery, or access to a malicious website.

2. Email Profile

Field	Observed value
File	000b3003b95fb4a634cadb2fde61cc74.eml
From	kckrisa@apex2000.net, dkshgkshdgh@yahoo.co.kr
To	james.javins@enron.com, john.singer@enron.com
Subject	Visual Help Pro v4.
Date	Mon, 23 Apr 2007 17:33:57 -0300
MIME structure	multipart/related → multipart/alternative → text/plain + text/html + image/gif
Embedded object	newspaper.gif (6,929 bytes)
Observed URL	http://j-chkmail.ensmp.fr (appears in X-Miltered header)
Authentication headers	No Authentication-Results, Received-SPF, DKIM-Signature or DMARC lines visible in supplied evidence
Observed DNS	yahoo.co.kr → 13.248.158.7 and 76.223.84.192; apex2000.net returned an SOA authority response but no A answer in the captured dig output

3. Findings and Analysis

Multiple From addresses [High]

The **From** field contains **two email addresses** from different domains: **apex2000.net** and **yahoo.co.kr**. No separate **Sender** header was found in the analyzed headers. This unusual structure matches the **Multiple From Header** technique documented in the EPVME dataset.

Email authentication evidence absent [Medium-High]

No **Authentication-Results**, SPF, DKIM, or DMARC results were found in the available email headers. Therefore, the analysis cannot confirm whether the email passed or failed these authentication checks.

MIME/HTML structure [Medium]

The email contains **text/plain**, **text/html**, and an embedded GIF image. The HTML uses a **Content-ID (CID)** reference for the image instead of a normal external image URL.

Embedded object [Low-Medium]

The embedded file **newspaper.gif** was extracted and confirmed as a valid **GIF89a image**. No executable attachment was identified in the analyzed email.

Sender/domain observations [Medium]

The sender addresses use **apex2000.net** and **yahoo.co.kr**. DNS and WHOIS information was checked during the analysis. The results provide additional context but **domain age or DNS information alone is not enough to determine whether an email is malicious**.

Body content [Medium]

The plain-text body contains fragmented and unrelated sentences. This unusual content may indicate a manipulated or generated message, but **the body content alone does not prove that the email is a phishing attempt**.

4. Authentication Review

No **Authentication-Results**, **Received-SPF**, **DKIM-Signature**, or DMARC results were found in the available email headers.

Therefore, the correct finding is:

Authentication evidence was not present in the captured email.

This does **not** mean that SPF, DKIM, or DMARC failed. It only means that there is not enough information in the captured email to confirm whether these authentication checks passed or failed.

DMARC checks whether the visible **From** address matches an authenticated SPF or DKIM identity. Since the required authentication results are not available here, **a positive or negative authentication result cannot be confirmed from this evidence alone**.

5. EPVME Dataset Correlation

The EPVME dataset contains different categories of email-based attacks, including **Multiple From Headers**, **From Header Injection**, **Empty Mail From**, **Subdomain Attacks**, **MIME Attacks**, and **Email XSS**.

In this sample, the **From** field contains **two different email addresses from different domains**. This structure is consistent with the **Multiple From Headers** category documented in the EPVME dataset.

The research publication associated with EPVME reports **14,266 samples** in the Multiple From Headers category.

Therefore, the dataset information provides **supporting evidence that the unusual From header is related to a known email manipulation technique**. The classification is based on the observed email characteristics and their correlation with this documented technique, rather than on the dataset label alone.

6. Indicators of Compromise / Evidence

Type	Indicator / Evidence	Assessment
Email file	000b3003b95fb4a634cadb2fde61cc74.eml	Primary evidence
Subject	Visual Help Pro v4.	Observed
From	kckrisa@apex2000.net; dkshgkshdgh@yahoo.co.kr	Multiple identities
Recipient	james.javins@enron.com; john.singer@enron.com	Observed
Domain	apex2000.net	Observed in From field
Domain	yahoo.co.kr	Observed in From field
IP	13.248.158.7	DNS result at time of analysis
IP	76.223.84.192	DNS result at time of analysis
Embedded file	newspaper.gif	Valid GIF89a image, 456 × 380 pixels, 6,929 bytes
SHA-256	517f43cbd0d2cfdd743e904dd5b78ff4189a5456e200363e15a368298019fbe3	SHA-256 calculated locally using sha256sum
URL	http://j-chkmail.ensmp.fr	Observed only in X-Miltered header
Technique	Multiple From header / identity confusion	Strong correlation with EPVME category

7. Analyst Workflow

1. Identified the **.eml** file and checked its basic file type and structure.
2. Reviewed the **sender, recipients, subject, date, and email headers**.
3. Checked the headers for available **SPF, DKIM, DMARC, and authentication information**.
4. Examined the **plain-text and HTML** parts of the email.
5. Checked for **URLs, HTML elements, and embedded content**.
6. Extracted the embedded **newspaper.gif** and performed **file-type, metadata, hash, strings, hex, and Binwalk analysis**.
7. Performed **WHOIS and DNS checks** on the observed domains.
8. Compared the unusual **From** header structure with the **documented Multiple From Header technique** in the EPVME dataset.

8. Final Assessment

Classification: MALICIOUS / PHISHING-RELATED SAMPLE

The email should be **treated as malicious within the scope of this analysis**. The main evidence includes:

1. **Multiple email addresses in the From field**, from different domains.
2. **No separate Sender field** was observed in the captured headers.
3. **No visible SPF, DKIM, or DMARC authentication results** were available in the analyzed message.
4. The email structure is **consistent with the Multiple From Header technique** documented in the EPVME dataset.

The embedded **newspaper.gif** was also analyzed separately. It was confirmed as a valid GIF image, and the static analysis did not identify an obvious malicious payload.

9. Proof of Concept (PoC)

The following screenshots are reproduced from the analyst-provided DOCX evidence. They are retained as supporting evidence and are not treated as independent external validation.

```
(kali@kali)-[~/Downloads]
$ file ./1/000b3003b95fb4a634cadb2fde61cc74.eml
./1/000b3003b95fb4a634cadb2fde61cc74.eml: news or mail, ASCII text
```

Evidence 1: supplied terminal capture

```
(kali@kali)-[~/Downloads]
$ grep -Ei "(From|To|Subject|Date|Reply-To|Return-Path|Received|Message-ID|Content-Type):" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
From:[kckrisa@apex2000.net], <dkshgkshdghh@yahoo.co.kr>
To: james.javins@enron.com, john.singer@enron.com
Subject: Visual Help Pro v4.
Date: Mon, 23 Apr 2007 17:33:57 -0300
Content-Type: multipart/related;
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: image/gif;
```

Evidence 2: supplied terminal capture

```
(kali@kali)-[~/Downloads]
$ grep -Ei "Authentication-Results|Received-SPF|DKIM-Signature|DMARC" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
(kali@kali)-[~/Downloads]
$ grep -i "spf" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
(kali@kali)-[~/Downloads]
$ grep -i "dkim" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
(kali@kali)-[~/Downloads]
$ grep -i "dmarc" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
```

Evidence 3: supplied terminal capture

```
(kali@kali)-[~/Downloads]
$ sed -n '1,15p' ./1/000b3003b95fb4a634cadb2fde61cc74.eml
From:[kckrisa@apex2000.net], <dkshgkshdghh@yahoo.co.kr>
To: james.javins@enron.com, john.singer@enron.com
Subject: Visual Help Pro v4.
Date: Mon, 23 Apr 2007 17:33:57 -0300
MIME-Version: 1.0
Content-Type: multipart/related;
type="multipart/alternative";
boundary="====_NextPart_000_0003_01C785CD.92B82500"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2800.1106
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2800.1106
X-Miltered: at mailchk-m02 with ID 462D18BB.001 by Joe's j-chkmail (http://j-chkmail.ensmp.fr)!
X-Virus-Scanned: ClamAV version 0.90.2, clamav-milter version 0.90.2 on localhost
X-Virus-Status: Clean
```

Evidence 4: supplied terminal capture

```

(kali@kali)-[~/Downloads]
└─$ python3 -<<PY
from email import policy
from email.parser import BytesParser

path = "/1/000b3003b95fb4a634cadb2fde61cc74.eml"

with open(path, "rb") as f:
    msg = BytesParser(policy=policy.default).parse(f)

print("From:", msg.get("From"))
print("Subject:", msg.get("Subject"))

for part in msg.walk():
    if part.get_content_type() in ("text/plain", "text/html"):
        print("\n---", part.get_content_type(), "---")
        print(part.get_content())

PY
From: <., dkshgkshdghh@yahoo.co.kr
Subject: Visual Help Pro v4.

--- text/plain ---
Your loving fingers wake in fragrant pages a History of a love which slept,. Identifies the type of class interface that is generated for a class.
Displays a help screen showing the options, special characters, and defaults for GREP.
Other problems have been acknowledged.
Such terminators should not be used with the DBI. Neither have many other methods, such as the organisation of entertainments, etc.
Maybe the dukhi left us some firewood.
The US covers a total area of 9,372,610 square kilometers and a total land area of 9,166,600 square kilometers.
Visual Help Pro v4.
A search is the only way to do it.
Displaying Long Strings and arrays of Const Strings.
The originating host must compose this option with a large enough timestamp data area to hold all the timestamp information expected. More likely both of us.

--- text/html ---
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.0 Transitional//EN">
<HTML><HEAD>
<META http-equiv=Content-Type content="text/html; charset=iso-8859-1">
<META content="MSHTML 6.00.2900.2180" name=GENERATOR>
<STYLE></STYLE>
</HEAD>
<BODY bgColor=#ffffff>
<DIV><IMG alt="" hspace=0 src="cid:000201c785e6$b8055d00$99cd0cbd@SERVERBSC"
align=left border=0></DIV>
<DIV><FONT face=Arial size=2>Your loving fingers wake in fragrant pages a
History of a love which slept,. Identifies the type of class interface that is
generated for a class.</FONT></DIV>
<DIV><FONT face=Arial size=2>Displays a help screen showing the options, special
characters, and defaults for GREP.</FONT></DIV>

```

Evidence 5: supplied terminal capture

```

(kali@kali)-[~/Downloads]
└─$ grep -Ei "<a |href=|src=|iframe|form|script" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
timestamp data area to hold all the timestamp information expected. More =
src=3D"cid:000201c785e6$b8055d00$99cd0cbd@SERVERBSC"=20
a large enough timestamp data area to hold all the timestamp information =

```

Evidence 6: supplied terminal capture

```

(kali@kali)-[~/Downloads]
└─$ grep -Eoi 'https?://[^\s">]+' ./1/000b3003b95fb4a634cadb2fde61cc74.eml
http://j-chkmail.ensmp.fr/

(kali@kali)-[~/Downloads]
└─$ grep -nEi "Content-type: image|Content-ID:|Content-Disposition:|filename=" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
101:Content-type: image/gif;
104:Content-ID: <000201c785e6$b8055d00$99cd0cbd@SERVERBSC>

```

Evidence 7: supplied terminal capture

```

(kali@kali)-[~/Downloads]
$ python3 - <<'PY'
from email import policy
from email.parser import BytesParser

path = "./1/000b3003b95fb4a634cadb2fde61cc74.eml"

with open(path, "rb") as f:
    msg = BytesParser(policy=policy.default).parse(f)

for part in msg.walk():
    if part.get_content_type() == "image/gif":
        filename = part.get_filename() or "embedded.gif"
        data = part.get_payload(decode=True)

        with open(filename, "wb") as out:
            out.write(data)

        print("Extracted:", filename)
        print("Size:", len(data), "bytes")
PY
Extracted: newspaper.gif
Size: 6929 bytes

```

Evidence 8: supplied terminal capture

```

Session Actions Edit View Help
(kali@kali)-[~/Downloads]
$ exiftool newspaper.gif
ExifTool Version Number      : 13.55
File Name                    : newspaper.gif
Directory                    : .
File Size                     : 6.9 kB
File Modification Date/Time   : 2026:08:26 02:45:00-04:00
File Access Date/Time        : 2026:08:26 02:45:09-04:00
File Inode Change Date/Time   : 2026:08:26 02:45:00-04:00
File Permissions              : -rw-rw-r--
File Type                    : GIF
File Type Extension          : gif
MIME Type                    : image/gif
GIF Version                  : 89a
Image Width                  : 456
Image Height                 : 380
Has Color Map                : Yes
Color Resolution Depth       : 1
Bits Per Pixel               : 8
Background Color             : 83
Duration                     : 486.55 s
Image Size                   : 456x380
Megapixels                   : 0.173

```

Evidence 9: supplied terminal capture


```

(kali@kali)-[~/Downloads]
$ whois apex2000.net | grep -Ei "Domain Name|Registrar|Creation Date|Updated Date|Expiry|Name Server|DNSSEC|Registrant"
Domain Name: APEX2000.NET
Updated Date: 2026-04-05T07:31:56Z
Creation Date: 2012-05-13T10:00:51Z
Registry Expiry Date: 2027-05-13T10:00:51Z
Registrar: CSL Computer Service Langenbach GmbH d/b/a joker.com
Name Server: NS1.APEXMIDLAND.NET
Name Server: NS2.APEXMIDLAND.NET
Name Server: NS3.APEXMIDLAND.NET
Name Server: NS4.APEXMIDLAND.NET
DNSSEC: unsigned

registrar's sponsorship of the domain name registration in the registry is
date of the domain name registrant's agreement with the sponsoring
automated except as reasonably necessary to register domain names or
about or related to a domain name registration record. VeriSign does not
domain names or modify existing registrations. VeriSign reserves the right
Domain Name: apex2000.net
Updated Date: 2026-04-05T07:31:56Z
Creation Date: 2012-05-13T10:00:51Z
Registrar: CSL Computer Service Langenbach GmbH d/b/a joker.com
Registrant Name: On behalf of apex2000.net
Registrant Organization: c/o IDPS International Domain Privacy Services GmbH
Registrant Street: Hansaallee 191
Registrant City: Duesseldorf
Registrant Postal Code: 40549
Registrant Country: DE
Registrant Phone: +49.21186767448
Registrant Fax: +49.211867676448
Registrant Email: https://cs1-registrar.com/contact/apex2000.net/owner
Name Server: ns1.apexmidland.net
Name Server: ns2.apexmidland.net
Name Server: ns3.apexmidland.net
Name Server: ns4.apexmidland.net
DNSSEC: unsigned

```

Evidence 10: supplied terminal capture

```

(kali@kali)-[~/Downloads]
$ dig apex2000.net

;<<<> DiG 9.20.23-1-Debian <<<> apex2000.net
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 14102
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: e85870c1dd27e5bd01000006a8e8b98a42eefe4380116c8 (good)
;; QUESTION SECTION:
;apex2000.net.                IN      A

;; AUTHORITY SECTION:
apex2000.net.                 3600    IN      SOA     ns1.apexmidland.net. support.cloudns.net. 2022083110 7200 1800 1209600 3600

;; Query time: 427 msec
;; SERVER: 192.168.18.1#53(192.168.18.1) (UDP)
;; WHEN: Wed Aug 26 02:45:44 EDT 2026
;; MSG SIZE rcvd: 143

(kali@kali)-[~/Downloads]
$ nslookup apex2000.net
Server:      192.168.18.1
Address:     192.168.18.1#53

Non-authoritative answer:
*** Can't find apex2000.net: No answer

```

Evidence 11: supplied terminal capture

```

(kali@kali)-[~/Downloads]
$ whois yahoo.co.kr | grep -Ei "Domain Name|Registrant|Registered Date|Last Updated Date|Expiration Date|Name Server|DNSSEC"
DNSSEC : unsigned
Domain Name : yahoo.co.kr
Registrant : Whois Corp.
Registered Date : 1997. 06. 04.
Last Updated Date : 2020. 03. 05.
Expiration Date : 2026. 10. 15.
DNSSEC : unsigned
Primary Name Server :
Secondary Name Server :

(kali@kali)-[~/Downloads]
$ dig yahoo.co.kr

;<>> DiG 9.20.23-1-Debian <>> yahoo.co.kr
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 11913
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 15ce7eddd4633e4c010000006a8e8bd903ec0f30c2e92234 (good)
;; QUESTION SECTION:
;yahoo.co.kr.                IN      A
;; ANSWER SECTION:
yahoo.co.kr.                 60      IN      A      76.223.84.192
yahoo.co.kr.                 60      IN      A      13.248.158.7

;; Query time: 251 msec
;; SERVER: 192.168.18.1#53(192.168.18.1) (UDP)
;; WHEN: Wed Aug 26 02:46:49 EDT 2026
;; MSG SIZE rcvd: 100

(kali@kali)-[~/Downloads]
$ nslookup yahoo.co.kr
Server:      192.168.18.1
Address:     192.168.18.1#53
Non-authoritative answer:
Name:   yahoo.co.kr
Address: 13.248.158.7
Name:   yahoo.co.kr
Address: 76.223.84.192

```

Evidence 12: supplied terminal capture

```

(kali@kali)-[~/Downloads]
$ grep -nEi "Content-Type: image|Content-ID:|Content-Disposition:|filename=" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
101:Content-Type: image/gif;
104:Content-ID: <000201c785e6$b8055d00$99cd0cbd@SERVERBSC>

(kali@kali)-[~/Downloads]
$ grep -Ei "href=" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

(kali@kali)-[~/Downloads]
$ grep -i "^X-Miltered:" ./1/000b3003b95fb4a634cadb2fde61cc74.eml
X-Miltered: at mailchk-m02 with ID 462D18BB.001 by Joe's j-chkmail (http://j-chkmail.ensmp.fr)!

(kali@kali)-[~/Downloads]
$ sed -n '95,115p' ./1/000b3003b95fb4a634cadb2fde61cc74.eml
<DIV><FONT face=3DArial size=3D2></FONT>&nbsp;</DIV>
<DIV><FONT face=3DArial size=3D2></FONT>&nbsp;</DIV></BODY></HTML>

-----=_NextPart_001_0004_01C785CD.92B82500--
-----=_NextPart_000_0003_01C785CD.92B82500
Content-Type: image/gif;
        name="newspaper.gif"
Content-Transfer-Encoding: base64
Content-ID: <000201c785e6$b8055d00$99cd0cbd@SERVERBSC>

R0lGODlhYAF8YvdTAAAMAI4ICAB3CIImDAAgKiXQ0iAB/hrbOwrfgsjpP9TgpDmkqAIQuDpcVC8KT
AuQgBwB3JACU3A9j1AftIDn9LAJuz8Bbc+ANE7AAxeACpjAEpUAFZaAHxpcCapWCCpjANhdAgCFABuG
AESLA2ztAH918KiGBp7Dt2EAACLBcyjAEGXAGWSC4ubDj2nAL2uAt6mAATFAB3NADu8AGLNA3a8
Aq69ALrDAejLDgnpACHuADPpAFLsAIPcAJbsAM7gA0XrAASGNB8KPkGAPFwAPYQHTakAO74MP90M
Tgs0ORsmOEwmPlk1S3cYQpUuShYSN9khPABDSiM6Njc5MWLJSn9N0a08SsVF0t1MNgBqMyxUNT1j
NGFrTnZoAKhWPc1iM9xnSQB0hZzPzF3QFd1ooF3MZp9PLSGPeeD0gCmPRWpNDi1jTlyZNYadRp+k
SbojPdSWtADNMRRCSk6yS217Nim/SJ/FP7fJRd3APQDJTBLgPEXkP2PsS4HZ05jVTLnU0PhNg0L
h1oGckQEhmEA1noAgpsAFLQAft8AfQwcdBYngEwXgmIWfIoW1pMsiCUXctIRhABJhx9FeEM4cWI1
dYUxeKhLdc50etFBhgBghhhngzhqf25qcYlmiKVdjrRW1+dYhw1IfCWfdEqKjmyMcneInq2Nhhp2
itqN1QCvciSzdKongltieYabf62We7mjJtetgwC1cS3NizHN1GS/h4HDd5031rfHf+vIhqjYec7d

(kali@kali)-[~/Downloads]
$ mousepad ./1/000b3003b95fb4a634cadb2fde61cc74.eml

```

Evidence 13: supplied terminal capture

```

(kali㉿kali)-[~/Downloads]
$ ls
1 EPVME_1.zip  FREECareerConsultation.eml  newspaper.gif

(kali㉿kali)-[~/Downloads]
$ sha256sum newspaper.gif
517f43cbd0d2cfdd743e904dd5b78ff4189a5456e200363e15a368298019fbe3  newspaper.gif

(kali㉿kali)-[~/Downloads]
$ file newspaper.gif
newspaper.gif: GIF image data, version 89a, 456 x 380

(kali㉿kali)-[~/Downloads]
$ exiftool newspaper.gif
ExifTool Version Number      : 13.55
File Name                    : newspaper.gif
Directory                   : .
File Size                    : 6.9 kB
File Modification Date/Time   : 2026:08:26 02:45:00-04:00
File Access Date/Time        : 2026:08:26 02:45:09-04:00
File Inode Change Date/Time   : 2026:08:26 02:45:00-04:00
File Permissions              : -rw-rw-r--
File Type                    : GIF
File Type Extension           : gif
MIME Type                    : image/gif
GIF Version                   : 89a
Image Width                   : 456
Image Height                  : 380
Has Color Map                 : Yes
Color Resolution Depth       : 1
Bits Per Pixel                : 8
Background Color              : 83
Duration                      : 486.55 s
Image Size                    : 456x380
Megapixels                    : 0.173

(kali㉿kali)-[~/Downloads]
$ xxd -l 32 newspaper.gif
00000000: 4749 4638 3961 c801 7c01 8753 0000 0c00  GIF89a .. | .. S ..
00000010: 8e08 0800 7708 8983 0008 0a89 740e 8800  ... .w ... .t ...

```

Evidence 14: supplied terminal capture

The following commands were used to examine the email, its headers, embedded content, and extracted GIF file.

1. file ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Identifies the file type and confirms that it is an email file.

2. grep -Ei "^(From|To|Subject|Date|Reply-To|Return-Path|Received|Message-ID|Content-Type):" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Extracts important email headers such as sender, recipient, subject, date, routing information, and MIME type.

3. grep -Ei "Authentication-Results|Received-SPF|DKIM-Signature|DMARC" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Checks whether authentication-related headers are present in the email.

4. grep -i "spf" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Searches the email for any SPF-related information.

5. grep -i "dkim" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Searches the email for any DKIM-related information or signatures.

6. grep -i "dmarc" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Searches the email for any DMARC-related information.

7. sed -n '1,15p' ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Displays the first 15 lines to inspect the beginning of the email and its headers.

8. python3 email parser script to inspect text/plain and text/html parts

Purpose: Separates and examines the plain-text and HTML portions of the email.

9. grep -Ei "\<a [href=|src=|iframe|form|script" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Searches for HTML links, external resources, iframes, forms, and scripts.

10. grep -Eoi 'https?://[^\<>]+' ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Extracts HTTP and HTTPS URLs found in the email.

11. grep -nEi "Content-Type: image|Content-ID:|Content-Disposition:|filename=" ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Identifies embedded images, Content-ID references, attachments, and filenames.

12. sed -n '95,115p' ./1/000b3003b95fb4a634cadb2fde61cc74.eml

Purpose: Displays the section containing the embedded content for closer inspection.

13. python3 email parser script to extract the image/gif attachment

Purpose: Extracts the embedded GIF from the email for separate file analysis.

14. file newspaper.gif

Purpose: Identifies the actual file type, format, and basic image properties.

15. sha256sum newspaper.gif

Purpose: Calculates the SHA-256 hash to uniquely identify the extracted file.

16. exiftool newspaper.gif

Purpose: Examines metadata and technical information stored in the GIF.

17. strings newspaper.gif | head -50

Purpose: Displays the first 50 readable strings found in the GIF.

18. strings newspaper.gif | grep -Ei 'http|https|script|javascript|cmd|powershell|exe|dll|base64'

Purpose: Searches the GIF for URLs, scripts, commands, executable references, or encoded-content indicators.

19. xxd -l 32 newspaper.gif

Purpose: Examines the first 32 bytes to verify the file signature and header structure.

20. xxd newspaper.gif | tail -20

Purpose: Examines the final bytes of the GIF and checks the file ending structure.

21. binwalk newspaper.gif

Purpose: Checks the GIF for embedded files, additional data, or unusual file signatures.

22. identify newspaper.gif

Purpose: Uses ImageMagick to verify the image format, dimensions, color information, and file size.

23. identify -verbose newspaper.gif | head -100

Purpose: Provides detailed ImageMagick information about the GIF, including image properties and statistics.

24. whois apex2000.net | grep -Ei "Domain Name|Registrar:|Creation Date|Updated Date|Expiry|Name Server|DNSSEC|Registrant"

Purpose: Retrieves available WHOIS registration and domain information for apex2000.net.

25. dig apex2000.net

Purpose: Performs a DNS lookup to examine DNS records associated with apex2000.net.

26. nslookup apex2000.net

Purpose: Performs an additional DNS lookup to verify the domain's DNS resolution.

27. whois yahoo.co.kr | grep -Ei "Domain Name|Registrant|Registered Date|Last Updated Date|Expiration Date|Name Server|DNSSEC"

Purpose: Retrieves available WHOIS registration and DNS-related information for yahoo.co.kr.

28. dig yahoo.co.kr

Purpose: Performs a DNS lookup to identify DNS records and IP addresses associated with yahoo.co.kr.

29. nslookup yahoo.co.kr

Purpose: Performs an additional DNS lookup to verify the domain's DNS resolution.

10. References

- **EPVME-Dataset GitHub repository** — <https://github.com/sunknighteric/EPVME-Dataset/>
- **EPVME_1.zip repository page** — https://github.com/sunknighteric/EPVME-Dataset/blob/main/EPVME_1.zip
- **A combined feature selection approach for malicious email detection based on a comprehensive email dataset** — <https://cybersecurity.springeropen.com/articles/10.1186/s42400-024-00309-6>
- **RFC 5322 — Internet Message Format** — <https://www.rfc-editor.org/rfc/rfc5322>
- **RFC 7489 — DMARC** — <https://www.rfc-editor.org/rfc/rfc7489>
- **CISA — Phishing guidance** — <https://www.cisa.gov/>